

 LETTER

REPLY TO WILLIG ET AL.:

Long-term population trends in the Luquillo Rainforest

Brad Lister^{a,1} and Andres Garcia^b

Temperatures

In our study (1), given Long-Term Ecological Research (LTER) information, we avoid data compromised by long-term averages and adjustments using data from other stations. Since temperatures for January 1975 to April 1978 include long-term averages, our data (1) begin in 1979. From 1979 on, we use temperatures corrected by the LTER with meteorological station data (1). Regarding Willig et al.'s (2) comment that we "combine data files that are not compatible to create the temperature record," we ask, What precisely does "not compatible" mean? What quantitative evidence exists—after correction and elimination of compromised data—that combined records are invalid? As support for the validity of our temperature data, we conducted a *t* test between mean annual temperatures for 1993 to 2015 at El Verde and Bisley (Table 1). The stations are a few kilometers apart, both at an elevation of 350 m. Hence, Bisley provides an independent check on the El Verde data. We previously show no significant difference in the slopes of temperature with time at the 2 sites (1), and there is also no difference in mean temperature (Table 2).

In their letter (2) and supplementary material (3), Willig et al. imply that combining "incompatible" data led to a spurious decrease in temperatures at El Verde. Changes in temperatures from 1998 to 2013 at El Verde and Bisley (Fig. 1) were actually part of a temporary global cooling (4). However, while mean maximum temperatures declined, mean minimum temperatures and days with temperatures >29 °C increased (1), suggesting that negative impacts of rising temperatures continued during the evening, as occurred in Panama (5).

Walking Sticks

We use the minimum number known alive for counts. Our downloaded data contain no entries for 1991. Data for 1992, 2012, 2013, and 2014 are not used, as they contain no dry-season data.

Arthropods

Willig et al. (3) state that canopy arthropods cannot be estimated because sampling of tree species was not random. This contradicts Schowalter (6): "Trees and branches were selected randomly." Hence, we assumed random sampling. To avoid confounding results, we included only taxa with data for all census periods. This excluded 54 taxa having data for just 2002 to 2009. A coccid, *Cryptolaemus montrouzieri*, had 1 day when numbers exceeded 1,200, ~20 times higher than all individuals from all other samples. This outlier was eliminated. Our regression of canopy arthropods vs. temperature is nonlinear (1); Willig et al. (3) use linear regression. In figure 5 A and B of ref. 1, the *y* axis should be labeled Log(Arthropods/FoliageWt+1).

Coquis

Concerning questions from ref. 3, all data came from the LTER. Willig et al. (3) also utilize unpublished data; hence, we cannot comment on differences with our results.

Birds

Willig et al. (3) state that we do not correct for variation in sampling effort ranging from 382.5 to 995.0 net-hours per year. The LTER website provides no information other than nets were run for 4 consecutive days per year. We understandably concluded that this was the standard effort.

Regressions

Taking the natural logarithm of the data (3) invalidates the assumption of Poisson regression that data are discrete counts. Hence, all of the Poisson regressions are wrong. Willig et al. (2, 3) fail to dislodge our conclusion that increasing temperatures are negatively impacting the fauna of the Luquillo forest. However, a more holistic understanding entails samples from more habitats and further analysis of the interaction of droughts, hurricanes, Atlantic Multi-decadal Oscillation, and El Niño/Southern Oscillation.

^aDepartment of Biological Sciences, Rensselaer Polytechnic Institute, Troy, NY 12180; and ^bEstación de Biología Chamela, Instituto de Biología, Universidad Nacional Autónoma de México, 47152 Chamela, Jalisco, Mexico

Author contributions: B.L. and A.G. designed research; B.L. and A.G. performed research; B.L. and A.G. analyzed data; and B.L. wrote the paper. The authors declare no conflict of interest.

Published under the [PNAS license](#).

¹To whom correspondence may be addressed. Email: listeb@rpi.edu.

Published online May 29, 2019.

Table 1. Results of an independent-samples t test for mean maximum annual temperatures taken between 1993 and 2015 at El Verde and Bisley

Location	Levene's test		t Test for equality of means				
	F	Significance	t	Degrees of freedom	Two-tailed significance	Mean difference	SE difference
El Verde–Bisley temperature equal variances	3.857	0.056	−1.423	42	0.162	−0.285	0.200
Unequal variances			−1.407	35.58	0.168	−0.285	0.202

Table 2. Mean maximum annual temperatures at El Verde and Bisley

Group	N	Mean temperature	SD	SEM
El Verde 1.0	21	26.92	0.747	0.163
Bisley 2.0	23	27.20	0.577	0.120

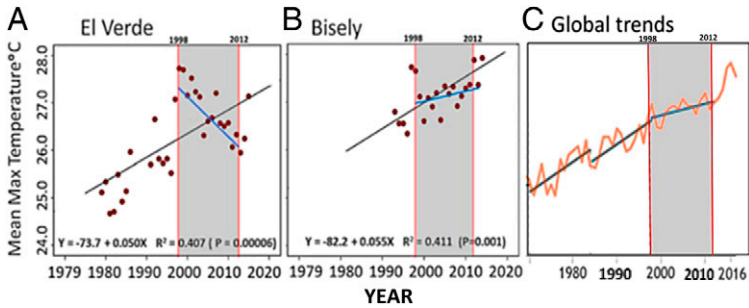


Fig. 1. (A and B) Blue lines indicate shifts in slope that began in 1998/1999 and ended by 2013. Reprinted with permission from ref. 1. (C) Global temperature trends showing the period of slower warming from 1998 to 2013. Reprinted with permission from ref. 4.

1 B. C. Lister, A. Garcia, Climate-driven declines in arthropod abundance restructure a rainforest food web. *Proc. Natl. Acad. Sci. U.S.A.* **115**, E10397–E10406 (2018).
2 M. R. Willig *et al.*, Populations are not declining and food webs are not collapsing at the Luquillo Experimental Forest. *Proc. Natl. Acad. Sci. U.S.A.* **116**, 12143–12144 (2019).
3 M. R. Willig *et al.*, Supplementary materials (2019). <https://luq.lter.network/pop-trends-yunque-luquillo>. Accessed 10 April 2019.
4 R. Lindsey, Did global warming stop in 1998? NOAA.GOV (2018). <https://www.climate.gov/news-features/climate-qa/did-global-warming-stop-1998>. Accessed 10 April 2019.
5 J. Stapley, M. Garcia, R. M. Andrews, Long-term data reveal a population decline of the tropical lizard *Anolis apletophallus*, and a negative affect of el nino years on population growth rate. *PLoS One* **10**, e0115450 (2015).
6 T. D. Schowalter, Invertebrate community structure and herbivory in a tropical rain forest canopy in Puerto Rico following Hurricane Hugo. *Biotropica* **26**, 312–319 (1994).